Penetration Testing Basic TASK-1

1.Install Nmap on your system and run a basic scan of your local network (use 'nmap -sn <your subnet>/24') to discover active devices. Save the output to a file named scan_results.txt.

Ans :

sudo apt update

sudo apt install -y nmap

ip -4 addr

nmap -sn 192.168.1.0/24 -oN scan_results.txt

cat scan_results.txt

2.Pick any popular app you use daily (like Instagram, Zomato, or Paytm) and write one paragraph describing what information a penetration tester might try to gather during the reconnaissance phase for that app.

Ans : For a popular app like Instagram, a penetration tester might gather information during the reconnaissance phase to understand the app's attack surface and identify potential security weaknesses. This information may include the official website and subdomains, IP addresses, domain registration details, publicly available APIs, technologies and frameworks used by the app, server information, and publicly accessible employee or developer profiles. The tester might also examine the app's login and registration pages, password recovery process, publicly available documentation, and mobile application permissions to understand how the system works. This reconnaissance helps identify potential entry points and areas that may require further security testing. All information gathering should be conducted within the scope of authorization provided by the app owner.

**3.Use Nmap to perform a port scan on a public test server (such as scanme.nmap.org) with the command 'nmap -sS scanme.nmap.org' and list all open ports you find in your submission.

Hint: Only scan approved test targets like scanme.nmap.org, never random IPs.**

Ans :

Port	State	Service
22/tcp	open	SSH
80/tcp	open	HTTP
9929/tcp	open	Nping-echo

4.After completing a scan, write a short report (5-7 lines) summarizing what you discovered, including the number of hosts found, open ports, and any potential risks you noticed. Format your report as if you were submitting it to a client.

Ans :

1. he Nmap scan was performed on the authorized test server scanme.nmap.org.
2. One target host was specified for the scan; its live status should be confirmed from the actual scan output.
3. Example open ports include 22 (SSH), 80 (HTTP), and 9929 (Nping-echo); these require verification against the scan results.
4. Open SSH ports may pose risks if weak passwords or outdated software are used.

5. HTTP traffic may be unencrypted, potentially exposing sensitive information.
6. Only services confirmed by the scan should be included in the final findings.